DSIS-DPR:Structured Instance Segmentation and Diffusion Prior Refinement for Dental Anatomy Learning

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***Abstract*—Instance segmentation in medical imaging plays a crucial role in clinical diagnostic tasks, and have shown promising performance in practical applications. In this paper, we discuss a more fine-grained instance segmentation task: dental structured instance segmentation based on panoramic radiographs. However, direct segmentation of tooth structures encounters inherent challenges. Traditional instance segmentation networks often fall short in capturing intricate internal features, and exacerbated by the frequent blurring found in medical imaging, which can result in the deficiency of anatomical details. To deal with these problems, we propose a novel framework called DSIS-DPR, which combines a dental structured instance segmentation (DSIS) network with an enhanced diffusion prior refinement (DPR) method. Specifically, our innovatively designed structure-aware network leverages fine-grained feature fusion, acquiring a richer representation of internal anatomical structures. With the integration of adversarial learning, the model is primed to deliver holistic and subtle predictions of tooth structures. Furthermore, taking inspiration from dentists' inherent ability to utilize prior knowledge, such as understanding dental structures to label invisible anatomical structures, we propose a diffusion inpainting to refine the results of DSIS without additional annotations. Equipped with built-in structure learning, DPR is capable of modifying anomalies within each predicted segmentation, resulting in a more robust and complete structured segmentation result. Meanwhile, we ensure rigorous oversight over the reconstruction of areas affected by abnormalities, ensuring that any introduced adjustments minimally disrupt the well-predicted structured segmentation results. Extensive experiments have demonstrated that our DSIS-DPR outperforms all existing classical instance segmentation networks. The collected dataset is available: https://github.com/Zzz512/TSD.**

***Index Terms*—Abnormal Detect, Diffusion Inpaint, Dental Panoramic Imaging, Instance Segmentation, Structured Segmentation**

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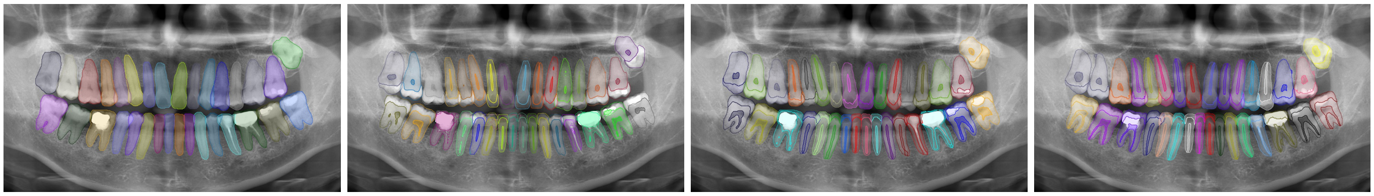
I. INTRODUCTION

DENTAL diseases, recognized as one of the prevalent chronic conditions [1], have been highlighted as potential precursors to systemic ailments such as cardiovascular diseases [2] and gastrointestinal tumors [3]. In clinical dental diagnosis, panoramic dental radiographs offer advantages including low radiation exposure, cost-effectiveness, and rapid imaging, establishing themselves as one of the paramount radiological diagnostic tools in dentistry. By instance segmentation, diagnostic systems can accurately detect targets that are centered around teeth in panoramic images. This enables a intact depiction of each tooth's morphology, dimensions, position, and classification. Several studies [4], [5] have successfully applied instance segmentation on panoramic images of dental medicine, demonstrating impressive performances. Although tooth instance segmentation in dental imaging provides valuable object-level information, its application has mainly been limited to preliminary tasks [6]–[8], such as adjacently separating, individual count, missing detection, or instances classification. In practical diagnostics, a thorough examination of major oral diseases often relies on the internal structures of each tooth and its surrounding environment. For example, the distance from the enamel-cementum junction to the bottom of the periodontal pocket provides clear indicators of periodontal disease. Furthermore, the depth of dental caries (shallow, moderate, or deep) is determined based on the lesion's extent [9], ranging from the outer surface of the tooth's dentin to its pulp. Existing instance segmentation methods face challenges when dealing with detailed and subtle diagnostic tasks. This emphasizes the pressing demand for advanced detection methods to enhance medical diagnosis.

To meet the specified requirements, we put forward a novel task called dental structured instance segmentation. This task goes beyond segmenting the entire tooth instance by also delineating its internal structures. Considering that each tooth comprises identical compositions, existing approach [10], [11] involve modification of the mask decoder head, transforming it into a multi-class from binary segmentation. In this paradigm, the network excels at handling the broader instance segmentation of the entire tooth, as shown in Fig. 1 (a). However, it falls short in capturing the intricate internal details of the tooth shown in Fig.1 (b). To enhance the inherent repre-



(a) Instance Segmentation **(b) Existing Method** **(c) DSIS w/o Refinement (ours)** **(d) DSIS-DPR (ours)**

Fig. 1. The overview of challenges and advancements in dental radiograph analysis are shown as follows: (a) Traditional instance segmentation networks provide merely basic instance-level insights. (b) When delving into fine-grained structured segmentation, many existing techniques fall short in comprehensively capturing the intrinsic structures of teeth. (c) In contrast, our Dental Structured Instance Segmentation (DSIS) method excels by offering a more profound understanding of these intricate details. (d) Taking it a step further, leveraging the Diffusion Prior Refinement (DPR) enables the rectification of previously anomaly intricacies, bringing our predictions more in alignment with real-world expectations.

sentation of instance details, we introduce a novel structure-aware instance segmentation framework named dental structured instance segmentation (DSIS), which is characterized by fusion compensation through an auxiliary encoder. DSIS extracts fine-grained features from the relevant input region of the instance and merges them with high-level semantic features in instance feature map, generating enhanced the internal feature representation of the instance. Moreover, DSIS incorporates adversarial learning [12] to guide the network towards predicting more comprehensive dental structures. By take the whole segmentation network as generator, we are able to obtain all tooth instances and then effectively capturing their fully complete internal structures, as depicted in Fig. 1 (c).

Whereas the network mentioned above can reveal instance intrinsic structures, there are still some tough problems. In medical imaging, visual representations are frequently characterized by indistinct features [13], including blurred boundaries and the inevitable loss of built-in internal structures [14], [15]. As a result, accurately segmenting the dental structures within defective image seem impossible. Fortunately, certain tissues, such as teeth, cells, or eyeballs, exhibit inherent anatomical compositions. We have observed that annotators possess a profound understanding of the definitive structures of specific instances, allowing them to use this prior knowledge to infer and complement disordered structures [16], even in invisible areas of instance. This insight inspires us to explore the reconstruction of all dental structure defects through prior knowledge learning. For the first time, we combine instance segmentation techniques with diffusion models, introducing the method of diffusion prior refinement (DPR), which exploits the diffusion model's capacity to learn the prior distribution of dental structures, assisting DSIS in rectifying aberrant predictions. Initially, we exclude individuals without pathological structures from the manual dental instance annotations. By trainning on these instances, the diffusion model captures the distribution of non-diseased dental instances only. This ensures that the refinement process exclusively generates regular dental structures, as shown in Fig. 1 (d). Subsequently, we propose a targeted dental structure inpainting mechanism based on the diffusion model. To ensure the accuracy of the regions in structural predictions, DPR follows a principle of minimal reconstruction. Hence, we meticulously design an anomaly detection module to identify areas in structural predictions that require refinement. Finally, DPR adjusts the predicted structural masks based on the detected anomaly masks, resulting in a comprehensive dental structure segmentation prediction.

In the completion of medical image segmentation using diffusion models [17], we observed that generative models, particularly those based on diffusion principles, exhibit distinct characteristics compared to unremarkable segmentation networks. These generative models demonstrate an ability to perceive the texture of images from coarse to fine, formulating more complete structural or morphological representations. In contrast, segmentation networks typically focus on strict pixel alignment between the inputs and outputs. However, structured segmentation often encounters defects due to the invisible or blurred representation of imaging features. In this context, the DPR module emerges as a corrective tool. It leverages prior learning of inherent structures to effectively modify any flaws predicted by the segmentation network. This capability is demonstrated in Figure 1, where the DPR module enhances the completeness and accuracy of the segmentation results. Our contributions can be summarised as follows:

1) We propose a novel framework for dental structured instance segmentation (DSIS) with effective segmentation of teeth while preserving their internal structures, through fine-grained feature fusion compensating for structural detail, along with adversarial learning to enhance the capacity of capturing dental structures.
2) By emulating how humans explore external learning to label ambiguous structures, we introduce diffusion prior refinement (DPR) for DSIS. Via leveraging prior knowledge of regular dental structures, we can refine abnormalities marked by a customized anomaly detector.
3) Extensive experimental results demonstrate that DSIS-DPR achieves state-of-the-art performance when compared to existing classical instance segmentation methods. This is particularly evident in its effectiveness in depicting challenging-to-detect pulp structures through the DPR mechanism.

II. Related Work

A. Instance Segmentation

Instance segmentation, encompassing both single-stage and two-stage frameworks, has given rise to numerous advanced network architectures. Mask R-CNN [18], which builds on object detection for instance segmentation, and its successors, the R-CNN series [19], [20], focus on extracting and segmenting regions of interest (RoI) [21] in images. Subsequently, He *et al.* [22] introduced PointRend, a method that calculates pixel values from selectively sampled irregular subsets of an image. To further exploit the symbiotic relationship between detection and segmentation, Chen *et al.* [23] and Fang *et al.* [24] introduced the hybrid task cascade (HTC) and shared QueryInst for instance segmentation, respectively. However, challenges persist regarding the disparity in the Intersection-over-Union (IoU) distribution of samples during training and inference. Addressing this, Thang *et al.* [25] proposed the sample consistency network (SCNet) to ensure a consistent IoU distribution between training and inference phases. DetectoRS [26] brought innovation with its "look and think twice" strategy, blending unique feature integration techniques. Structured segmentation techniques play a crucial role in image analysis [27], [28], particularly in the anatomical examination of human tissues. For efficient cell and nucleus segmentation, Bouyssoux *et al.* [29] introduced dedicated methods. In demonstrating the versatility of structured segmentation, Shu *et al.* [30] and Boutillon *et al.* [31] highlighted its applications in cardiac and musculoskeletal anatomy, respectively. Whereas segmentation methodologies have seen substantial advancements, they still encounter challenges, especially when internal instance information is required, which is crucial for intricate medical diagnoses.

B. Diffusion Prior Model

In recent years, diffusion models, inherently characterized by Markov processes, have emerged as a dominant paradigm for image generation and synthesis [32], [33]. Notably, their performance has surpassed that of traditional generative frameworks such as the Variational Autoencoder (VAE) and Generative Adversarial Networks (GAN). In the context of clinical medicine, ensuring the fidelity and veracity of radiological presentations is paramount. Thus, the integration of diffusion models into medical image analysis has become particularly noteworthy. These models have been primarily employed for tasks including image denoising, high-resolution reconstructions, and cross-modal synthesis [34], demonstrating their proficiency in learning the prior distribution of various imaging modalities. One of the most compelling characteristics of diffusion models is their inherent high-dimensional generation, laying the groundwork for more sophisticated endeavors. An intriguing avenue of application involves integrating conditional control networks with diffusion models [35]. Such characteristics hold the potential to precisely navigate the diffusion process or produce designated outcomes. Additionally, it introduces possibilities in the domain of restoring or reconstructing obscured regions [36] throughout the diffusion process. However, a significant challenge in the medical image reconstruction field is the potential risk of losing invaluable information when inpainting the original image. Many existing diffusion models lack the robust measures necessary, resulting in an inability to meticulously oversee reconstructed regions. This limitation emphasizes the pressing need for further research and innovation in the area, with the aim of upholding both the authenticity and quality of reconstructed images.

III. Proposed Method

The proposed methodology is depicted in Fig.2, which can be summarized in three principal modules. Firstly, we detail our DSIS, a purpose-built network designed for accurate instance segmentation of dental structures in panoramic images. DSIS incorporates fine-grained features from the original images with high-level semantic features, resulting in a comprehensive representation of the internal structures of each instance. To enhance the predictive quality, an adversarial training strategy is implemented to distinguish between the real and predictive structural masks. Subsequently, we formulate an anomaly detection module to identify potential abnormalities of the predictions. The anomaly detection reviews predictive inconsistencies through a pseudo label generator during training and aggregating various detected anomaly maps into anomaly alpha masks at inference phase. Finally, by taking the anomaly alphpa mask as condition, we modify both the predicted structures and original dental instances with DPR. Here, the meticulous inpaint process ensures the preservation of both structure and overall image consistency.

A. Dental Structured Instance Segmentation

Considering the accomplishments of previous works in effectively detecting and segmenting tooth instances based on dental panoramic images, we introduce a versatile detection backbone that consists of a ResNet [37] and a Feature Pyramid Network (FPN) [38] to precisely localize each tooth instance. The backbone ranks all identified instances and selects the top N based on its corresponding scores as candidates for structured segmentation. As same as general two-stage instance segmentation network, we extract Regions of Interest (RoI) from the feature maps, generating features that describe individual tooth. We take the f_i^S to represent *i*-th instance features derived from the FPN as high-level semantic features.

The success of instance segmentation networks can be attributed to the equivalent size matching between instance segmentation mask and the detected bbox. However, for structured segmentation, some dental structures (such as enamel or pulp) of an instance tend to be much smaller than the size of bbox. As a result, the detection semantic features may not fully capture the information of dental internal structures. Therefore, we employ an auxiliary encoder E that carries out a secondary feature extraction from the region cropped from the original image according to the bbox, capturing image-level features f_i^I for instance i. To compensate a more fine-grained image-level feature f_i^I into the semantic feature f_i^S, we interpolate f_i^S by setting it to twice the size of the semantic feature map. Finally, fuse and decode are conducted to get the predicted structure masks $\hat{s}_i$ as follows:

$$\hat{s}_i = \mathcal{M}\left(f_i^I + \mathcal{T}\left(f_i^S\right)\right), \tag{1}$$

where $\mathcal{M}$ denotes a transpose convolution and $\mathcal{T}$ denotes the mask decoder consists of several non-scaling convolution layers. And the final convolution layer performs multi-class prediction for each pixel, leading to a structured instance mask of instance. Structured instance segmentation loss is defined as multi-class cross entropy loss.

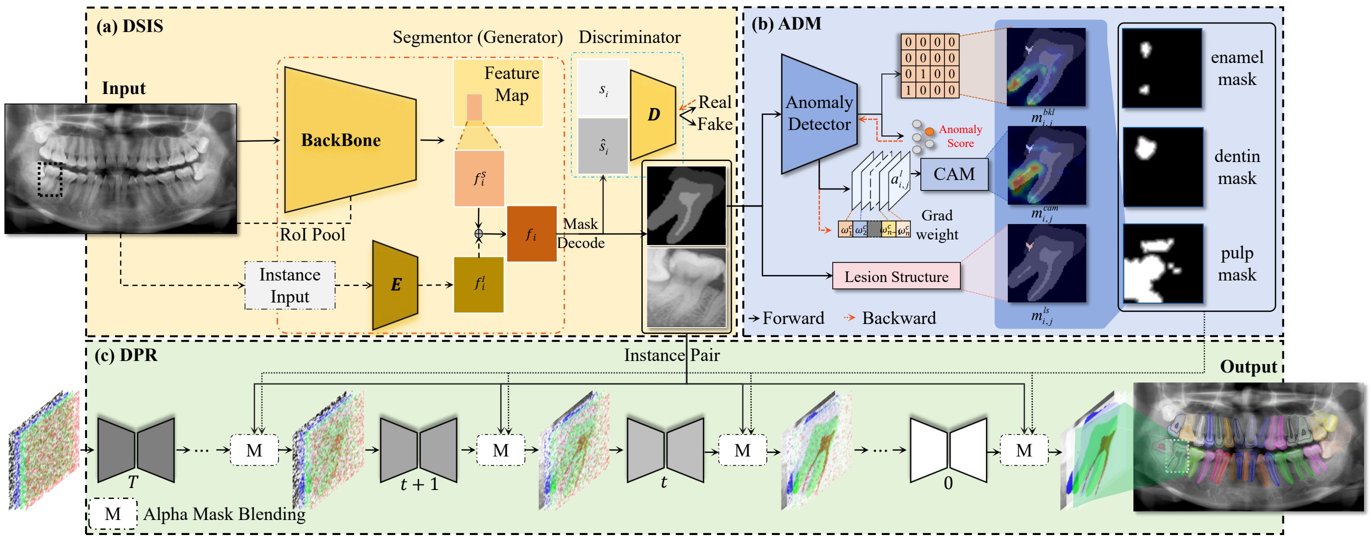


Fig. 2. The pipeline of our proposed DSIS-DPR. Where the dental panoramic image is first submitted to Dental Structured Instance Segmentation (DSIS) network. The resultant prediction inspected by an Abnormality Detection Module (ADM). Subsequently, Diffusion prior refinement (DPR) modifies the predicted structural mask and the image based on the anomaly alpha mask.

One notable challenge in the results of structured instance segmentation is the presence of imperfections in certain subtle structures. It becomes easy to discern these predictive defects in the structure mask due to discrepancies between some predicted dental structures and the manual annotations. Adversarial learning for the segmentation provides an unsupervised method to improve predicted structure mask quality, which allows us to refine the prediction masks of dental structured instance segmentation. A discriminator $\mathbf{D}$ is introduced to distinguish between the predicted structured masks $\hat{s}_i$ and the annotated structured masks s_i. Given the remarkable differences in the distributions of these two masks, the discriminator can easily identify defective structure masks. In iterative learning, the entire structured segmentation network is regarded as the generator $\mathbf{G}$, and then we optimize the adversarial loss as follows:

$$\ell_{dsc}(s_i, \hat{s}_i; \mathbf{\Theta_D}) = -\log(\mathbf{D}(s_i)) - \log(\mathbf{D}(1 - \hat{s}_i)), \quad (2)$$

$$\ell_{adv}(I, x_i^I; \mathbf{\Theta_G}) = -\log(\mathbf{D}(\mathbf{G}(I, x_i^I))), \quad (3)$$

where $\mathbf{\Theta_G}$ and $\mathbf{\Theta_D}$ are the parameters of the segmentation network and discriminator, respectively. I denotes the input dental panoramic image, while x_i^I represents the tooth instance cropped from I based on the detected bbox. By jointly optimizing (2) and (3), DSIS is inclined to come into being a more comprehensive and authentic dental structure predictions.

Considering that the structured segmentation in DSIS can accurately predict the masks for most structures, it implies that not every instance requires calibration when the DPR algorithm is executed. It is only necessary to correct structures that are identified as incomplete. However, due to the stochastic nature of the DPR sampling process, there is a possibility that areas correctly predicted by DSIS might be erroneously altered post-refinement, leading to issues of over-refinement or under-refinement. Therefore, the algorithm should adhere to the principle of minimal reconstruction.

B. Anomaly Detection Module

Considering that the structured segmentation in DSIS can accurately predict the masks for most structures, it implies that not every instance requires calibration when the DPR algorithm is executed. It is only necessary to correct structures that are identified as incomplete. However, due to the stochastic nature of the DPR sampling process, there is a possibility that areas correctly predicted by DSIS might be erroneously altered post-refinement, leading to issues of over-refinement or under-refinement. Therefore, the algorithm should adhere to the principle of minimal reconstruction. Our anomaly detection module begins with a pseudo-label generator that directly checks the integrity of the predictive structural masks. Taking into account that the structure of healthy teeth are always continuous and smooth, any structure with multiple separated mask regions is marked as an anomalous structural class. Thus, we examine the connectivity of the j^{th} structure over the predicted mask for the i^{th} instance, represented as $\hat{s}_{i,j}$. Here, j does not exceed the total dental structural categories, K (i.e., $j \leq K$). If $\hat{s}_{i,j}$ contains more than two connected domains (given that a regular structure has only foreground and background), we set the anomaly class $y_{i,j}^c$ to 1; otherwise, it is set to 0. For a given structure, it is represented by a binary mask where 1 indicates the foreground region of the instance structure, and 0 represents the background. A normal dental structure would contain only one connected foreground region. The presence of two or more connected regions, resulting from segmentation of the connected area, indicates that the structure has been compromised, marking it as an abnormal structure class. The anomaly class supervised learning helps to identify structural breaks for each predictive mask during inference. Nevertheless, there are some structural deficits or redundancies without evident structural breaks that can avoid the integrity checking. To precisely detect these fields, we employ a block-level discrimination by referring to VQGAN [39]. Anomaly detector slices $\hat{s}_{i,j}$ and $s_{i,j}$ into

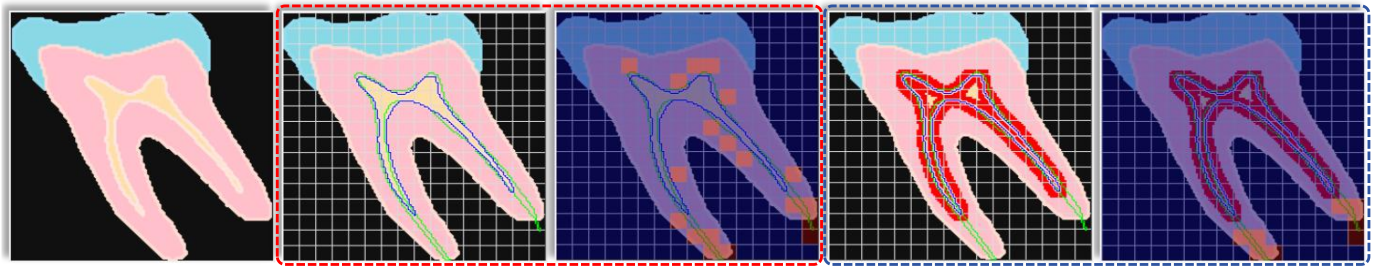


Fig. 3. The process of generating pseudo-labels for anomalous pulp structure blocks. (a) is the predicted structure mask put into the anomaly detector. (b) illustrates the direct generation of anomaly block pseudo-labels based on the annotated pulp structure (green contour) and the predicted pulp structure (blue contour) with the anomalies shown in the high-heat sections of (c) where the pseudo-label is True. (d) and (e) demonstrate the generation of anomaly block pseudo-labels after scaling, with the red areas being suppressed.

non-overlapping blocks, yielding block anomaly maps $s_{i,j,m,n}$ and $\hat{s}_{i,j,m,n}$, respectively, where (m,n) indicates the position of the sliced block in the maps. We make up anomaly block pseudo-label as follows:

$$y^{\text{blk}}_{i,j,m,n} = \begin{cases} 1 & \text{if } \mathbb{I}\left(j \in s_{i,m,n} \wedge j \notin s'_{i,m,n}\right) \\ & +\mathbb{I}\left(j \notin s_{i,m,n} \wedge j \in s'_{i,m,n}\right) > 0 \\ 0 & \text{otherwise} \end{cases}, \tag{4}$$

where $\mathbb{I}(\cdot)$ is the indicator function, which is scheduled to be 1 if the conditions in the parentheses are true, and 0 otherwise. Since the predictive mask $\hat{s}_i$ does not fully align with the annotated mask s_i, the pseudo-label generator tends to set the block-level pseudo-label $y^{blk}_{i,j}$ to 1 at the edges of the predicted structure. We thus take scaling, as illustrated in Fig.3, to combat this problem. Now, the detector receives the predicted structure mask $\hat{s}_i$, producing anomalous class $\hat{y}^c_{i,j}$ and block-level maps $\hat{y}^{blk}_{i,j,m,n}$. In the end, the pseudo-label allows us to train abnormality detection by cross-entropy loss.

With detected anomaly results $\hat{y}^c_{i,j}$ and $\hat{y}^{blk}_{i,j}$, we extract both the lesion structure areas as well as the abnormal structure regions. The lesion maps m^{ls}_i are directly obtained from $\hat{s}_i$ according to the predicted pathological structure. While the abnormal block prediction results $\hat{y}^{blk}_{i,j}$ are upsampled to compose the abnormal block areas $m^{blk}_{i,j}$. For the anomaly class, we introduce the gradient class activation map (Grad-CAM) [40] to find the suspicious areas. Concretely, in the intermediate layer l of the detector, we put forward the predicted structure mask $\hat{s}_{i,j}$ for getting activation maps of layer l, denoted as $a^l_{i,j} \in \mathbb{R}^{C\times H\times W}$. We then require a gradient weight of l^{th} layer from specific anomaly class scores $\hat{y}^c_{i,j}$:

$$w^l = \frac{1}{Z}\sum_u\sum_v \frac{\partial \hat{y}^l_{i,j}}{\partial a^l_{i,j,u,v}}, \tag{5}$$

where u and v are the two-dimensional spatial position of $a^l_{i,j}$. Z represents the total pixel count in a feature spatial map, and $w^l \in \mathbb{R}^C$ is the class activation weight vector. By summing $a^l_{i,j}$ weighted by w^l, we have the class activation map of cracked abnormal structure, denoted as $m^{cam}_{i,j} \in \mathbb{R}^{H\times W}$:

$$m^{cam}_{i,j} = \text{Relu}\left(w^l \cdot a^l_{i,j}\right). \tag{6}$$

By aggregating $\left\{m^{ls}_{i,j}, m^{blk}_{i,j}, m^{cam}_{i,j}\right\}$, abnormality detector explores the complete anomaly structure mask $m_{i,j}$, revealing all anomaly areas of the j^{th} predicted structure over the i^{th} instance. It is used to guide structural repair as well as image reconstruction. Finally, to mitigate the effect of diffusion on the anomalous structure boundary [41], we conduct a fuzzing on $m_{i,j}$ to formulate an alpha mask for the inpaint of DPR.

C. Diffusion Prior Refinement

To ensure the effectiveness of the training process for diffusion models, it is crucial that each data instance used does not contain any abnormal structures or lesions. By doing so, the diffusion model will learn only the prior distribution of normal anatomical structures. Consequently, the sampling process during the restoration phase, which draws from this prior distribution, will not produce structures with defects. This approach thus guarantees the achievement of our objective: the complete restoration of intact dental structures. In DPR section, we simultaneously modify the predictive structural mask and the original instance based on the alpha mask. Initially, we pick out all the instances of teeth without any lesion area to build a regular dental instance set, $X = \{x_i, i = 1, 2, ..., N\}$. In particular, we concatenate the original instance image x^I_i with the separated predicted structure $\hat{s}_i$, as depicted in Fig.4, thus we have concatenated image-structure image denoted as $x_i \in \mathbb{R}^{(K+1),h,w}$. For prior perception, the diffusion model defines the forward and reverse sampling process as two Markov processes:

$$q(x^{1:T}_{i,j} \mid x^0_{i,j}) = \prod_{t=1}^{T} q(x^t_{i,j} \mid x^{t-1}_{i,j}), \tag{7}$$

$$p_\theta(x^{0:T}_{i,j}) = p(x^T_{i,j})\prod_{t=1}^{T} p_\theta(x^{t-1}_{i,j} \mid x^t_{i,j}), \tag{8}$$

where j indexes the specific channel of concatenated image to be refined. $x^0_{i,j}$ and $x^T_{i,j}$ are final sampled images and pure noise sampled from Gaussian distribution $\mathcal{N}$, respectively. The forward sampling $q(x^t_{i,j} \mid x^{t-1}_{i,j})$ and the posterior probability $p_\theta(x^{t-1}_{i,j} \mid x^t_{i,j})$ are described as:

$$q(x^t_{i,j} \mid x^{t-1}_{i,j}) = \mathcal{N}(x^t_{i,j}; \sqrt{1-\beta_t}x^{t-1}_{i,j}, \beta_t \mathrm{I}), \tag{9}$$

$$p_\theta(x_{i,j}^{t-1} \mid s_{i,j}^t) = \mathcal{N}(x_{i,j}^{t-1}; \mu_\theta(x_{i,j}^t, t), \sigma_t^2 \mathrm{I}), \tag{10}$$

where β_t is the noise regulatory factor. Since DPR is exclusively trained on non-diseased teeth images and their associated structures, meaning refinement can only generate appearance of regular dental instances. Let $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{i=1}^t \alpha_i$. We can directly sample the noise image $x_{i,j}^t$ at step t based on the original concatenated image $x_{i,j}^0$ as follows:

$$q(x_{i,j}^t \mid x_{i,j}^0) = \sqrt{\bar{\alpha}_t} \cdot x_{i,j}^0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon, \tag{11}$$

where ϵ represents noise sampled from a standard Gaussian distribution. According to the DDPM, we have following optimization objective for the prior model:

$$\ell_{dm} = \mathbb{E}_{x_{i,j}^0, \epsilon, t \sim [1,T]} \left[\left\| \epsilon - \varepsilon_\theta \left(x_{i,j}^t, t \right) \right\|^2 \right], \tag{12}$$

where $\varepsilon_\theta(\cdot, t)$ is a noise prediction network with parameters θ, typically implemented by Unet, which predicts the potential noise for the sampled image at time step t. To refine dental image-structure pairs, we take a Gaussian sampling process for each denoising step. The process involves sampling $x_{i,j}^{t-1}$ from $x_{i,j}^t$ based on the probability distribution conditioned on ε_θ, which is presented as follows:

$$x_{i,j}^{t-1} \sim \mathcal{N} \left(\widetilde{\mu}_\theta(x_{i,j}^t, t), \Sigma_\theta(t) \right), \tag{13}$$

$$\widetilde{\mu}_\theta(x_{i,j}^t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_{i,j}^t - \frac{1 - \alpha_t}{1 - \bar{\alpha}_t} \varepsilon_\theta(x_{i,j}^t, t) \right), \tag{14}$$

$$\Sigma_\theta(t) = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t. \tag{15}$$

DPR applies the prior diffusion model trained on the dataset X to repair the predicted anomalous parts of $\hat{x}_{i,j}$. Unlike general inpaint methods based on diffusion models, they simply reconstruct image parts according to a single channel mask. Here, we use a separate alpha mask to blend with instance-separated image-structure pairs at each time step, allowing us to modify specific anomalous regions without affecting others. As illustrated in Fig.4, we assign different dental instance image along with structures to various channels. At the t^{th} time step of the sampling process, we obtain noise refined image as follows:

$$\hat{x}_{i,j}^{t-1,bld} = (1 - m_{i,j}) \cdot \hat{x}_{i,j}^{t-1,k} + m_{i,j} \cdot \hat{x}_{i,j}^{t-1,uk}, \tag{16}$$

where, $x_{i,j}^{t,bld}$ represents blending images. $\hat{x}_{i,j}^{t-1,k}$ is certainly known images directly sampled from $q(\hat{x}_{i,j}^{t-1})$ and $\hat{x}_{i,j}^{t-1,uk}$ is unknown images sampled from $p_\theta(\hat{x}_{i,j}^{t-1,uk} \mid \hat{x}_{i,j}^{t,bld})$, respectively. It is worth noting that our $m_{i,j}$ serves as an alpha mask that is capable of soft synthesizing the certainly know images and the sampled unknown images at the boundary regions. This effectively addresses disharmonies mentioned in [41] that can occur during the diffusion-based inpainting process. Refinement starts with a random noise $\hat{x}_{i,j}^T$ from standard normal distribution. A denoised image is then sampled according to (13). Subsequently, this is fused with the alpha mask following (16) for reconstruction. It is iterate until $t = 0$, at which stage the refined predictive structural mask $\hat{x}_{i,j}^0$, denoted as $\hat{x}'_{i,j}$, is obtained. $\hat{x}'_{i,0}$ and $\hat{x}'_{i,1:K}$ become the refined result of dental image and structured segmentation prediction for the i^{th} instance, respectively. Due to the randomness of the diffusion inpaint operation, the repaired structure mask may still contain abnormality. Fortunately, we find that DPR could iteratively repeat the operation of anomaly detection and diffusion prior refinement. It continues until a pre-defined number of repetitions is reached, or no further anomalies are detected, that is, $\sum_j m_{i,j} = 0$.

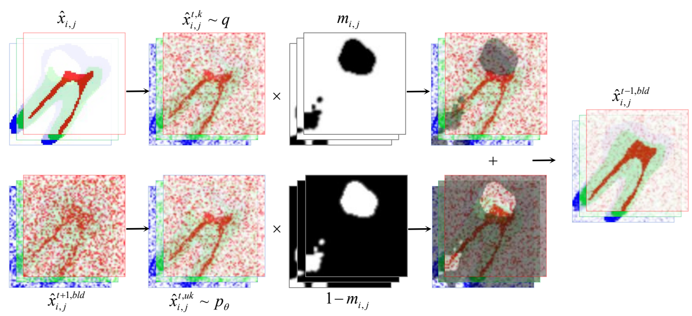


Fig. 4. Our separable inpaint synthesis mechanism employs the alpha mask blending for repairs. For regular regions, images are directly sampled from the original input, while for abnormal regions, images are sampled from the last iterative blend.

IV. Experimental Results and Discussions

A. Dataset Construction

In this study, we collect structured annotated instance segmentation data based on dental panoramic radiographs from 996 patients, covering a total 29112 instances for structured segmentation, which are split into 23365, 2850, and 2897 of training, validation, and test datasets, respectively. We meticulously label each dental instance by precise structural details of interior, specifically focusing on three intrinsic structures ($K = 3$), including enamel, dentin, and the pulp. Additionally, any individuals with residual roots, caries, or fractures are masked as decay areas, while prosthetic crowns, dental caps, and restorative fields are stamped and marked with filling materials mask, gathering in a dataset comprising 2,896 aberrant tooth instances. All panoramic images are rescaled to a resolution of $1,875 \times 1,044$.

B. Experiment Configurations

In our DSIS, we explore a two-stage architecture for dental structured instance segmentation. The network produce a 32×32, 256-channel instance semantic feature map and a resized 64×64 image feature map from the original radiographs via RoI extraction. Then the feature maps are fused, and handled by 4 convolutional layers of mask decoder. The discriminator of DSIS and anomaly detector of abnormality detection module take similar net configurations, with the anomaly block map set to 16×16. Both the anomaly classifier and abnormal block detector share an identical encoder, where the block output generated by three additional downsampling convolution operations. For DPR, a denoised U-Net processes concatenated images of the instance and three dental structures. Consequently, the input channel of ε_θ is 4 at a 64×64

TABLE I
COMPARISON OF VARIOUS INSTANCE SEGMENTATION METHODS ON THE SPECIFIED METRICS.

Method	instance			enamel			dentin			pulp		
	dc	asd↓	hd95↓	dc	asd↓	hd95↓	dc	asd↓	hd95↓	dc	asd↓	hd95↓
InstaBoost [42]	88.20	8.04	26.53	74.71	6.69	18.72	79.66	6.64	23.45	49.01	3.44	55.49
QueryInst [20]	85.55	8.57	24.80	64.56	22.22	37.74	76.78	6.62	21.91	53.49	6.58	37.32
HTC [24]	91.14	9.07	28.20	75.45	5.86	16.97	82.88	4.07	15.47	55.78	4.50	48.40
PointRend [22]	91.61	9.06	26.27	77.75	5.70	16.22	83.59	6.20	18.98	55.40	4.14	50.01
MS R-CNN [20]	90.59	9.08	27.50	75.95	5.73	15.14	83.15	4.99	16.85	59.96	4.07	41.32
DetectoRS [26]	91.95	8.82	27.50	77.15	5.12	14.58	84.19	4.17	14.58	60.32	3.61	40.31
Mask R-CNN [18]	92.44	8.55	26.72	78.99	4.93	14.79	85.46	4.27	13.96	61.26	3.95	29.43
Cascade R-CNN [19]	93.38	6.90	22.01	78.48	5.21	15.05	**86.30**	3.61	12.64	62.64	3.67	28.85
SCNet [25]	92.78	8.99	26.92	78.64	4.92	15.14	86.00	4.94	13.86	63.56	3.94	27.30
DSIS	92.95	3.77	12.93	77.99	5.00	15.23	84.41	4.25	14.25	63.90	3.89	28.83
DSIS-DPR	**93.59**	**3.30**	**11.50**	**79.14**	**4.48**	**14.39**	85.58	**3.60**	**11.76**	**70.29**	**3.33**	**13.49**

resolution. While time step is set to 1000, and β_t increases linearly from 1×10^{-4} to 0.02. We adopt adam optimizer with learning rates of 1×10^{-3} for DSIS and anomaly detection, but 1×10^{-5} for DPR's prior learning.

C. Evaluation Metrics

In our experiments, we employ the Dice Coefficient (DC), Average Surface Distance (ASD), and the 95% Hausdorff Distance (HD95) as evaluation metrics. These metrics are widely acknowledged in the field of image segmentation, particularly in medical imaging. The formulations of these metrics are presented as follows:

Dice Coefficient (DC) is defined by the equation:

$$\mathrm{DC} = \frac{2 \times |X \cap Y|}{|X| + |Y|}, \tag{17}$$

where X and Y represent the pixel sets in the predicted and ground truth segmentations, respectively. The term $|X \cap Y|$ indicates the number of pixels in the intersection of these two sets, whereas $|X|$ and $|Y|$ refer to the total number of pixels in each set. Higher values of the Dice Coefficient indicate better segmentation performance.

Average Surface Distance (ASD) is calculated using the equation:

$$\mathrm{ASD} = \frac{1}{N} \left(\sum_{x \in X_s} \min_{y \in Y_s} d(x,y) + \sum_{y \in Y_s} \min_{x \in X_s} d(y,x) \right), \tag{18}$$

here, X_s and Y_s denote the sets of surface pixels for the predicted and ground truth, respectively. N denotes $|X_s|+|Y_s|$. The function $d(x,y)$ measures the Euclidean distance between points x and y. ASD evaluates the average distance between the two surfaces, with lower values indicating more accurate segmentation.

95th percentile Hausdorff Distance (HD95) is determined by first calculating the set of distances $\{d(x,y)|x \in X_s, y \in Y_s\}$ and $\{d(y,x)|y \in Y_s, x \in X_s\}$, and then selecting the 95th percentile of this combined set as the HD95 value. This metric assesses the maximum distance between two sets of surfaces, with HD95 serving as a more robust variant by considering the 95th percentile of all distances to reduce the influence of outliers. Lower HD95 values reflect better segmentation quality.

D. Results of Proposed Method

Following Table I, our DSIS demonstrates superior performance in comparison with several commonly used instance segmentation networks. Considering that these methods are not readily adaptable for structured instance segmentation, we redefine their mask decoders in a manner consistent with DSIS, which allows them to predict multi-class mask within instances. We select three measurements of segmentation to measure innate enamel, dentin, and pulp structures accuracy. In addition, we aggregate these structures to formulate instance predictions simultaneously. With a specifically designed structure-aware module, our DSIS is able to confront the performance of all meticulous instance segmentation methods. In particular, our method demonstrates a notable improvement in the prediction of instance edges and overall performance for pulp structures. This is evident from the decrease in the asd of instances from 6.9 to 3.77, and the increase of dc of pulp from 61.26 (Mask R-CNN based) to 63.9, indicating a much closer alignment between the predicted boundaries and the structured GT contours. In addition, the dice score for pulp structures also exhibits remarkable enhancement. Notably, with the introduction of DPR, the model achieves top-tier performance over nearly all structures and metrics. Even the challenging recognition of pulp structure shows a 6.39% improvement in dice coefficient, accompanied by a 6.39 point reduction in hd95. Fig.5 reports the limitations of general instance segmentation methods in structured segmentation and highlights the proficiency of our DSIS-DPR in accurately segmenting intricate pulp structures, even in the case of pathological instances. Specifically, the first case illustrates the invisibility of the pulp chamber apex caused by overlapping alveolar bones, a problem nearly undetected by existing methods due to its lack of visual representation. DSIS can identify this latent structure to some extent, but in conjunction with DPR, it can fully excavate the entire dental pulp canal piercing through the dentin. The second case highlights the issue of structure disruption due to artificial filling materials, where existing methods result in significantly coarse segmentation. DSIS achieves smoother outcomes, whereas DPR can completely reconstruct the compromised dental structure. The final case shows the challenge of incomplete structural integrity due to crown fractures, where most methods fail to recognize the

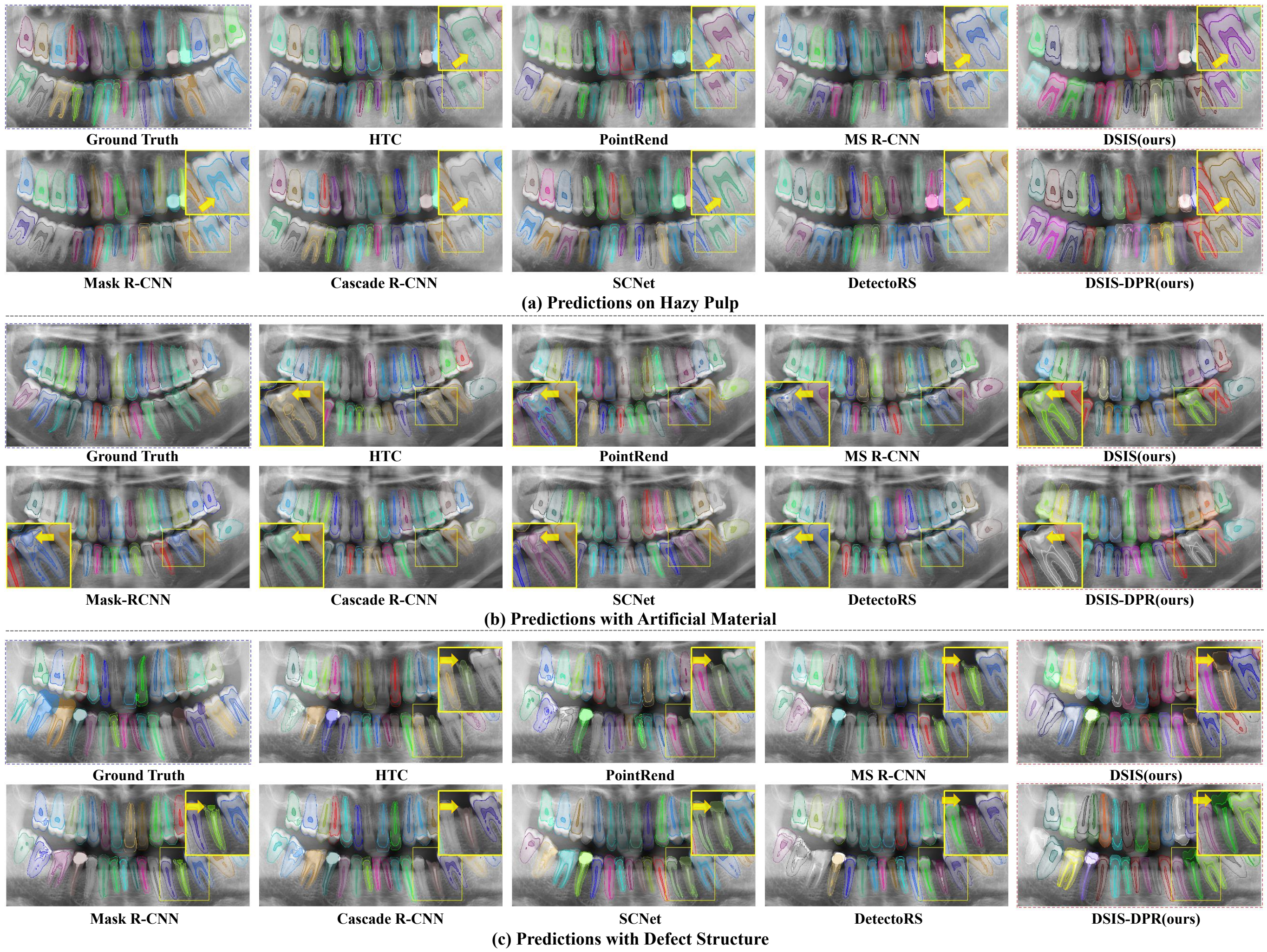


Fig. 5. visualization results of different DIS network and our proposed DSIS-DPR. On the whole, compared with common structured instance segmentation methods, our DSIS always tends to give a more complete instance structure. In detail, we show the predicted results of tooth structure in three various scenarios: (a) hazy pulp structure, (b) structure with manual repair material, and (c) structure with defects.

TABLE II
ABLATION EXPERIMENT OF STRUCTURED INSTANCE SEGMENTATION NETWORK

f_i^I	f_i^S	ℓ_{adv}	instance			enamel			dentin			pulp		
			dc	asd↓	hd95↓	dc	asd↓	hd95↓	dc	asd↓	hd95↓	dc	asd↓	hd95↓
			88.56	8.83	27.35	72.28	6.77	20.45	79.08	8.18	26.57	58.01	7.07	42.44
✓			87.10	7.88	26.39	60.94	18.56	64.70	78.29	6.34	23.91	39.80	6.34	61.87
	✓		89.79	6.82	21.64	72.31	6.08	18.64	80.82	6.28	20.88	58.26	5.18	37.66
	✓	✓	89.22	8.71	26.72	73.65	5.76	18.50	81.06	7.82	25.72	61.93	5.63	37.60
✓	✓		92.50	4.23	14.22	77.66	5.19	**14.90**	83.87	4.62	15.04	63.16	3.96	31.13
✓	✓	✓	**92.95**	**3.77**	**12.93**	**77.99**	**5.00**	15.23	**84.41**	**4.25**	**14.25**	**63.90**	**3.89**	**28.83**

missing part. DSIS is capable of completing the fragmented instance edges, while DPR can fully restore the structural details of the lost segments.

E. Ablation Study of DSIS

Here, we discuss compensated fusion strategies with different features received by mask decoders, and the effects of adversarial learning on structured segmentation. As shown in TABLE II, we assess the impact of three factors: f_i^I, f_i^S, and ℓ_{adv} for instance segmentation of three substructures enamel, dentin, and pulp. The table shows utilizing only image feature we have a relatively poor performance, indicating which can not properly represent dental structures alone. But incorporating both f_i^I and f_i^S together without ℓ_{adv} showcases a drastic enhancement, especially in reducing the asd and hd95 metrics across all structures, highlighting the effectiveness of feature fusion for compensating instance internal features boundary. Upon comparing the third and fourth rows of Table III, we observe that the introduction of the ℓ_{adv}, with only f_i^S active results in a decrease in instance dc from 89.79% to 89.22%. However, for the pulp structure, the dice increases 3.67%, and the asd improves from 5.18 to 5.63, signifying that adversarial loss helps to formulate fine intrinsic features. The employment of all three factors, f_i^I, f_i^S, and ℓ_{adv}, result reports in the best performance over nearly all evaluation metrics. Specifically, for the instance segmentation, the dice coefficient reach its peak at 92.95%, the asd is minimized to 3.77, and the hd95 comes to its lowest at 12.93. Similarly, for the substructures, most of the metrics observed their best performance when all factors activated. The aforementioned results underscore the availability of our proposed structure-aware module of DSIS for implementing superior structured instance segmentation.

TABLE III
ABLATION EXPERIMENT OF DIFFERENT ANOMALY DETECTOR SCALING

Scaling	instance		enamel		dentin		pulp	
	dc	hd95↓	dc	hd95↓	dc	hd95↓	dc	hd95↓
S=0	93.01	12.68	79.55	14.33	84.81	12.28	68.57	15.39
S=1	93.26	11.99	79.39	15.35	85.05	12.19	69.01	15.39
S=2	93.42	11.89	**79.68**	**14.32**	85.22	12.03	69.62	15.43
S=3	**93.59**	**11.50**	79.14	14.39	**85.58**	**11.76**	**70.29**	**13.49**
S=5	93.46	11.66	78.43	14.74	85.40	11.78	68.88	14.92

F. Quality Evaluation of Anomaly Detection

Given the predicted mask of the DSIS, the anomaly detection module aims to identify areas in the mask that need to be repaired. However, the pseudo-label generator is tend to rise a large number of false positive labels on the edge of the predicted structure mask, which is easy to cause the anomaly detection module to assign the edges as abnormal fields, bringing in unnecessary modification. With scaling, We mitigated this affect successfully through setting different scaling size to find the best performance of the method. In Table III, the effect of varying the scales (*S*) of the anomaly detector on the model's performance is laid out. An evident trend emerges as the scaling value increases: there's a general enhancement in the model's metrics up to *S*=3. Specifically, for instance segmentation, the dice coefficient sees an incremental increase from 93.01 to 93.59 at *S*=3. Similarly, the hd95 metric optimally decreases as *S* grows, reaching its best representation at *S*=3 for all structures. This trend underscores the significance of the factor (*S*) in refining the anomaly detection capability. However, a pivotal point seems to occur at *S*=3, pushing the scaling further to 5 shows a slight drop in all metrics. The diminished effectiveness, especially evident in the enamel structure's dc score and others, signals the potential pitfalls of over-scaling. We believe that increasing the scaling factor can enhance the detector's sensitivity for inconsistent block, but over-scaling may dilute critical contrasts during block pseudo-label generation, impacting the detector's accuracy. In essence, fine-tuning the scaling factor is essential to provide an exact anomaly area for DPR.

G. Diffusion Prior Refinement Process

To more clearly show the effectiveness of our novel proposed DPR, we visualize the refinement process of the modification operation, as depicted in Fig. 6. The structured instance segmentation network identifies the dental instances and acquires their anatomical structure, yielding predicted structural masks (third row). Subsequently, the anomaly detector assesses these predicted masks for anomalies, with detected abnormal regions highlighted using heatmaps (fourth row). For instances without pathological structures, refinements are rised directly based on the anomaly masks for specific structures only. Otherwise, the diffusion model simultaneously refines both the image and the structure, which follows the guidance by the lesion structure mask as well as the anomaly mask. Through the modification, the rectified image and structure align consistently, as depicted in last four columns of Fig. 6.

TABLE IV
DPR PERFORMANCE OF VARIOUS INPUT CHANNELS FORMULATION

Channels	instance		enamel		dentin		pulp	
	dc	hd95↓	dc	hd95↓	dc	hd95↓	dc	hd95↓
w/o DPR	92.95	12.93	77.99	15.23	84.41	14.25	63.90	28.83
1×	92.33	13.59	75.72	34.31	84.50	13.56	70.18	15.28
3×	93.01	12.74	78.26	14.68	84.80	12.31	70.21	13.84
4×	93.59	11.50	79.14	14.39	85.58	11.76	70.29	13.49

Fig.7 (a) depicts incorrect modification under aggregated DPR and that further improved by the separated structure blending (b), where the structures are departed into various channels for diffusion. In the early explorations, we employ single channel (1×), separated channels without instance image guidance (3×), and combined channels with instance image (4×) for refinement, resulting in the performance of DPR as shown in TABLE IV. The results clearly indicate that all refinements lead to a substantial improvement in the segmentation of intricate pulp structures. Nevertheless, when using DPR with a single channel, it results in a reduction of dice scores for both instance and enamel by 0.72% and 2.27%, respectively. It also doubles the hd95 of enamel, signifying that the single-channel refinement adversely affects the accuracy of predictions for other structures. By separating structural channels (3×), all structures can be reconstructed alone without affecting each other. Moreover, the combined channels (4×) yield superior performance, revealing that instance image-guided DPR ensures the consistency of segmentation.

The success of our DSIS-DPR primarily hinges on the diffusion's assembly generation capability, wherein instance images merge with occluded predicted masks. A primary concern for DPR is to produce consistent masks that match the anatomical structure of detected instance images. To validate this, we randomly sample multiple groups, with each containing 350 fully generated image-structure pairs. We evaluated the Structural Similarity Index Measure (SSIM) and Peak Signal-to-Noise Ratio (PSNR) scores between the generated images and the real dataset, as detailed in TABLE V. Additionally, we conduct manual annotations of the generated image structures to ascertain their accuracy. Overall, each tooth's generated

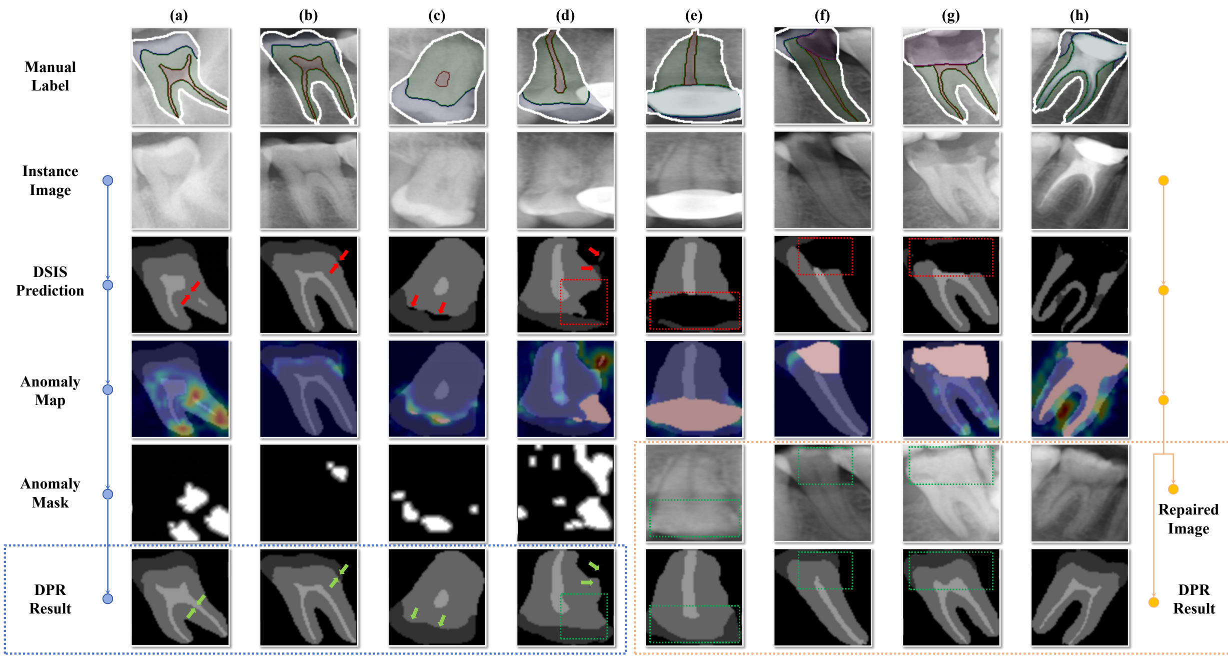


Fig. 6. The process of the diffusion prior refinement. We report several prevalent cases with perturbed even destroyed dental structure: (a) pulp imperceptible under high exposure, (b) regions overlapped with adjacent instance, (c) enamel-dentin demarcation, (d) indiscernible boundaries and interference from neighboring tooth treatments, (e) restored crowns, (f) dental lesions, (g) broken tooth, and (h) root canal filling materials. For all these cases, our DPR is capable of capturing their innate structural forms and coming up with a comprehensive structured predictions.

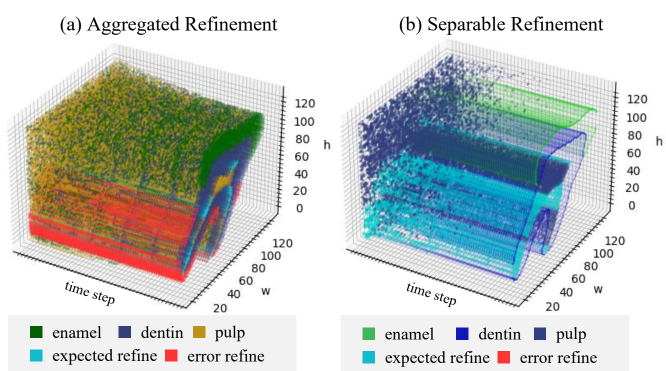


Fig. 7. Comparison of diffusion blending process for pulp. Where (a) reports aggregated blending introducing error refinement out of expected areas. By separating structures (b), DPR will never affect both well-predicted enamel and dentin structures. This approach ensures that the regular structure remains uncompromised during modification.

structure demonstrate a commendable degree of accuracy. This suggests that the reconstructed region of separated structure inpainting approach meet our anticipations. Moreover, we find a consistent dice score of roughly 70% for the pulp across all groups. This score closely matches the DPR outcome post the majority of dental pulp restorations, emphasizing our effective employment of DPR for pulp structure adaptations.

TABLE V
CONSISTENCY ASSESSMENT OF REFINED INSTANCE STRUCTURES

Group	SSIM↑	PSNR↑	instance	enamel	dentin	pulp
#1	0.775	24.37	95.44	80.02	88.13	71.91
#2	0.741	22.69	95.15	82.14	88.00	70.45
#3	0.710	21.59	95.09	84.24	87.89	70.21
#4	0.623	20.07	95.02	86.14	87.78	69.33

H. Iterative Diffusion Prior Refinement

We have discovered that the DPR (Diffusion Prior Refinement) module can be effectively reused in an iterative fashion to achieve superior structural reconstruction results. After the initial correction by the DPR, while most of the abnormal or lesioned structures are addressed, some unrealistic reconstructed structures may remain, or some minor details might be overlooked. By re-evaluating the repaired structure masks for anomalies and reprocessing them through the DPR, it is possible to identify and correct anomalies that were either not detected during the first correction or emerged as a result of it, thereby achieving enhanced structured segmentation. As demonstrated in Fig. 8, during the reconstruction of the pulp chamber structures in two instances, some inconsistencies at the tips, such as the pulp not breaking through the dentin or the disintegration of the pulp structure, are observed. A second iteration can yield a more accurate segmentation of the pulp. After the second correction, the structure is essentially complete, and in most cases, there is no need for a third or further iterations.

I. Efficient Discussion

The structural segmentation capabilities of our model, it also compromises the immediacy of instance segmentation due to the iterative nature of the diffusion model's sampling process. Fortunately, several strategies exist to mitigate this issue. For instance, adopting techniques such as Denoising Diffusion Implicit Models (DDIM) or model quantization can

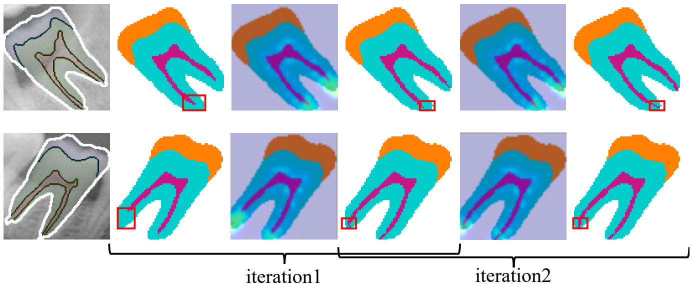


Fig. 8. The iterative repair results for two instances are visualized and arranged in two rows. After completing the first round of iterative corrections, some undesired structures still emerge. Through a second correction, these uncorrected or newly emerged inconsistencies are addressed.

markedly accelerate inference. Additionally, our approach does not necessitate adjustments for every individual instance; typically, only a minority of predictions present issues, meaning that the computational demand for correcting a few instances is manageable. Importantly, given the specific context of medical imaging, where the premium is placed on accuracy over real-time performance, we posit that the trade-off of some efficiency for a substantial increase in precision through DPR integration is warranted.

V. Conclusion

In this study, we focus on structured instance segmentation of teeth using dental panoramic radiographs, with the goal of simultaneously delineating tooth instances and their internal structures. To address the limitations of existing instance segmentation networks in adequately capturing intra-instance structural features, we introduce a strategy that combines feature fusion compensation and adversarial learning to improve the network's ability to perceive fine structural details. Confronting challenges such as lesions or image pervasively blurring in the dental panoramic images, which result in structural degradation or absence, we suggest a diffusion prior refinement operation to acquire prior knowledge about the dental structures previously. Drawing inspiration from the human annotation process where external knowledge is introduced, we expect to apply a prior model to learn actual information about tooth structure from manual annotations. This prior model subsequently guides the structure segmentation network to complete these predictive lesion areas, enabling us to obtain more comprehensive segmentation results for tooth instance. So far, experimental findings further reveal that our DPR possesses profound structural insights. The current approach modifies instance segmentation through the diffusion prior model, which is constrained by an anomaly detection mask. By integrating a more sensitive anomaly detector, we anticipate that structured instance segmentation could be optimized to achieve even better performance.

Fundamentally, our research explores two distinct approaches to instance segmentation: the discriminant model and the generative model. The discriminant model, exemplified by segmentation methods, learns a direct mapping from images to masks. However, this approach often becomes mired in minutiae, leading to failures in scenarios with ambiguous representations. In contrast, the generative model employs the AIGC (Artificial Intelligence Guided Context) technique, which captures the specific distribution of a given context. This enhances its resistance to interference across datasets. However, it also introduces a slower and more stochastic process due to the complex calculations involved in sampling. At the core of our work is the quest to balance efficiency and accuracy in structured instance segmentation. We propose that further investigation into the trade-offs between diffusion models and segmentation networks could significantly advance fine-grained instance segmentation.

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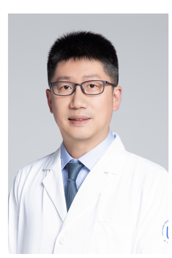

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